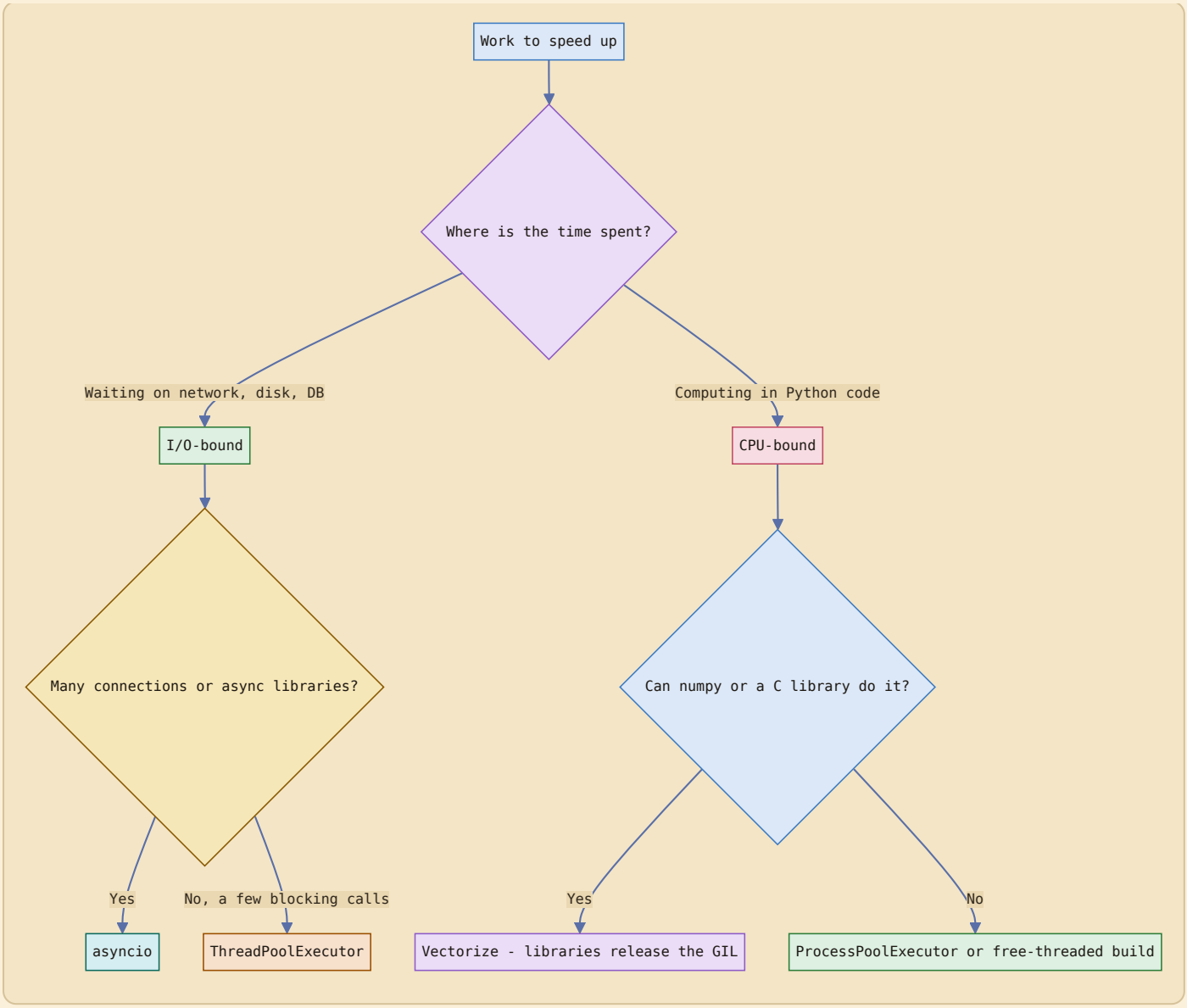
Chapter 16 — Concurrency and Async

> **What you will learn.** The difference between concurrency and parallelism, how to tell I/O-bound work from CPU-bound work, and Python's four main tools for doing several things at once: **threads** (`threading`, locks, `queue.Queue`), **processes** (`multiprocessing`), the high-level **executors** (`concurrent.futures`) and **async/await** (`asyncio`: event loop, coroutines, tasks, `gather`, `TaskGroup`, timeouts, async iteration). You will also learn what the **GIL** is, why it exists, and where the **free-threaded** (no-GIL) CPython stands in 3.13 and 3.14. The chapter ends with a "which tool?" table and the common pitfalls: blocking calls inside async code, race conditions and deadlocks.
>
> **Prerequisites.** Functions and closures (Ch 7), generators (Ch 11), context managers (Ch 12).
>
> **Code.** `chapters/16-concurrency-and-async/code/` — `threads_basics.py`, `gil_benchmark.py` (makes the plot in `img/`), `multiprocessing_demo.py`, `futures_demo.py`, `asyncio_demo.py`, `pitfalls_demo.py`. All outputs below come from Python 3.14.7 (default build, GIL on) on a 14-core Mac. Timings will differ on your machine; the *shape* of the results will not.
>
> **Java side.** The general theory — memory models, `synchronized`, `ReentrantLock`, semaphores, latches, atomics, concurrent collections, deadlock/livelock/starvation and the classic LLD problems — is in the LLD Handbook, Chapter 12 — Concurrency concepts. This chapter covers what is specific to Python.

Words used everywhere in this chapter (defined once):

Term	Meaning
Process	A running program with its own memory space. Two processes do not see each other's variables.
Thread	A path of execution *inside* a process. Threads of one process share its memory.
Task (asyncio)	A coroutine scheduled on an event loop. Many tasks share one thread.
Blocking call	A call that makes the caller wait (reading a socket, `time.sleep`, a disk read).
Race condition	A bug where the result depends on the unlucky timing of two threads touching shared data.
Critical section	Code that must be run by only one thread at a time.
Lock / mutex	An object that lets only one holder into a critical section.
Deadlock	Two or more threads each waiting forever for a lock the other holds.
GIL	Global Interpreter Lock: in standard CPython, only one thread runs Python bytecode at a time.



16.1 Concurrency vs parallelism

What it is

- **Concurrency** = *dealing with* many things at once: several tasks are in progress and their steps interleave. It can happen on a single CPU core.
- **Parallelism** = *doing* many things at the same instant: tasks literally run simultaneously on different cores.

Analogy: one chef cooking three dishes — putting pasta on to boil, then chopping while it boils — is concurrency. Three chefs each cooking one dish is parallelism. Concurrency is about *structure* (not sitting idle while waiting); parallelism is about *execution* (more hands).

Why it matters

Picking the wrong model wastes effort. Adding threads to a CPU-heavy pure-Python loop in standard CPython does nothing (the GIL allows one thread at a time), while adding threads to a program that spends 95 % of its time waiting on HTTP responses can make it 10–50× faster.

How it works

	Concurrency	Parallelism
Needs multiple cores?	No	Yes
Goal	Stay busy while waiting	Finish computation sooner
Python tools	threads, asyncio	multiprocessing, free-threaded threads, C extensions that release the GIL
Typical workload	Web servers, scrapers, chat apps	Image processing, simulations, number crunching

TL;DR

Concurrency overlaps waiting; parallelism overlaps computing. Parallelism is a kind of concurrency, but not the other way round.

16.2 I/O-bound vs CPU-bound

What it is

- **I/O-bound**: the program mostly *waits* — for the network, disk, a database, a user. The CPU is idle during the wait. Example: downloading 100 web pages.
- **CPU-bound**: the program mostly *computes*. The CPU is at 100 %. Example: computing SHA-256 of a billion strings in a Python loop, resizing images in pure Python.

How to tell (3 quick checks)

1. **Watch CPU usage** (`top`, Activity Monitor). ~100 % of one core → CPU-bound. Low CPU but slow → I/O-bound.
2. **Profile** (Ch 17): if the top functions are `socket.recv`, `select`, `read` — I/O. If they are your own loops — CPU.
3. **Compare wall time and CPU time**: `time.perf_counter()` (wall clock) vs `time.process_time()` (CPU time used). If wall ≫ CPU, you are waiting.

Measured

`gil_benchmark.py` runs 4 CPU tasks (a pure-Python loop of 15 million iterations) and 4 I/O tasks (`time.sleep(0.5)`, standing in for a network wait) three ways:

```python
def cpu_task(n: int = 15_000_000) -> int:
    total = 0
    for i in range(n):                    # pure-Python loop: holds the GIL the whole time
        total += i * i
    return total

def io_task(_: int = 0) -> None:
    time.sleep(0.5)                       # sleeping releases the GIL, like waiting on a socket
```

Output:

```
Python 3.14.7, GIL enabled: True
CPU-bound: sequential 1.94s  threads 1.83s  processes 0.87s
I/O-bound: sequential 2.02s  threads 0.51s  processes 0.92s
saved gil_benchmark.png
```

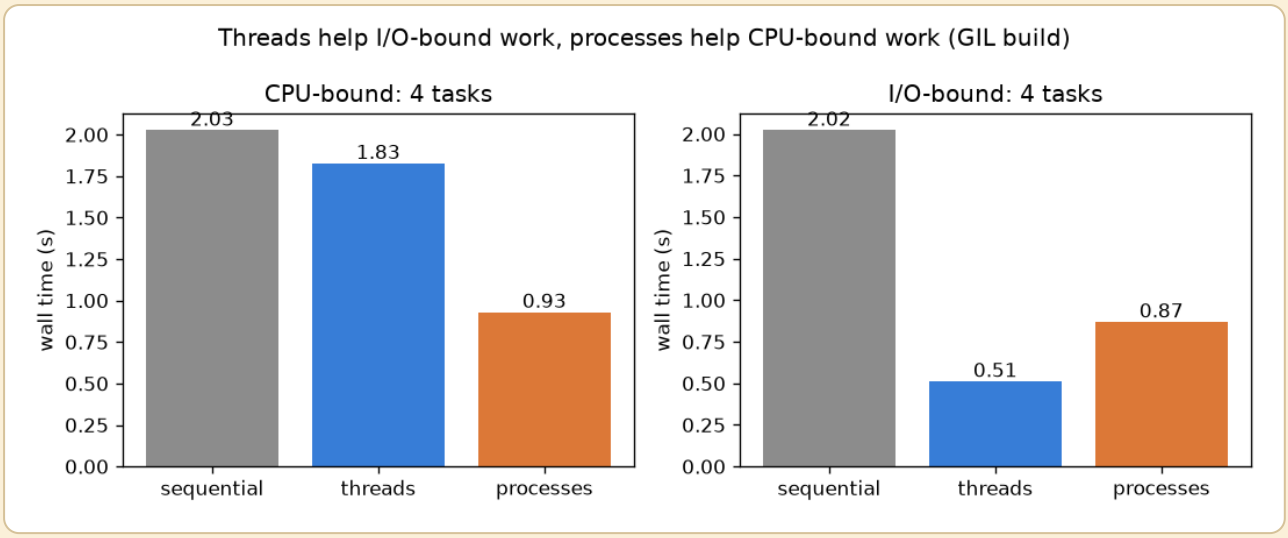


Figure — CPU-bound work only speeds up with processes; I/O-bound work speeds up with threads (processes also help but pay ~0.3–0.4 s start-up cost).

What it shows:

- **CPU-bound + threads ≈ sequential** (1.83 s vs 1.94 s is noise-level): the GIL lets only one thread run Python code at a time.
- **CPU-bound + processes ≈ 2.2× faster**: each process has its own interpreter and its own GIL, so they run on separate cores. It is not 4× because starting processes (the `spawn` method re-imports the module) and sending data back costs time.
- **I/O-bound + threads ≈ 4× faster**: while one thread sleeps (waits) it releases the GIL, so all four waits overlap.

TL;DR

Measure first. Waiting → threads or asyncio. Computing → processes, vectorized libraries, or (new) the free-threaded build.

16.3 Threads with `threading`

What it is

The `threading` module starts extra threads inside your process. All threads share global variables and objects on the heap. Each thread has its own call stack.

How it works

1. Create: `t = threading.Thread(target=func, args=(...), name="T1")`.
2. `t.start()` — the OS starts a new thread that calls `func(*args)`.
3. `t.join()` — the caller waits until `t` finishes (optionally `join(timeout)`).
4. A **daemon** thread (`daemon=True`) is killed abruptly when the main program exits; a normal thread keeps the program alive until it finishes. Use daemons only for work that can be lost (a heartbeat), never for writing files.

```python
def worker(name: str, delay: float, results: dict) -> None:
    time.sleep(delay)                    # stands in for waiting on I/O
    results[name] = threading.current_thread().name

threads = [threading.Thread(target=worker, args=(f"job{i}", 0.2, results), name=f"T{i}")
           for i in range(3)]
for t in threads: t.start()              # all three begin waiting at once
for t in threads: t.join()               # wait for all to finish
```

Output (`threads_basics.py`, line 1):

```
3 x 0.2 s sleeps in threads took 0.21 s -> {'job0': 'T0', 'job1': 'T1', 'job2': 'T2'}
```

Three 0.2 s waits took 0.21 s in total because they overlapped. Threads return nothing from `target`; to get results, write to a shared structure (as here), use a `queue.Queue`, or — better — use `ThreadPoolExecutor` (16.7), which gives you return values and exceptions.

An alternative style is subclassing `threading.Thread` and overriding `run()`; it works, but passing a `target` function is simpler and more common.

Per-thread data. Sometimes each thread needs its *own* copy of a value (a DB connection, the current request ID). `threading.local()` gives an object whose attributes are separate per thread. In asyncio code, where many tasks share one thread, use `contextvars.ContextVar` instead — each task sees its own value (web frameworks store the current request this way).

TL;DR

`Thread(target=..., args=...)`, `start()`, `join()`. Threads share memory — which is both their convenience and their danger.

16.4 Race conditions and locks

What it is

A **race condition** happens when two threads read-modify-write the same data and their steps interleave. Even `counter += 1` is not one step: CPython executes it as *load counter → add 1 → store counter*. If thread A loads 5, then thread B loads 5, both store 6 — one increment is lost.

A **lock** (`threading.Lock`) fixes it: only one thread at a time can hold it, so the read-modify-write becomes a **critical section** that cannot interleave.

The race, demonstrated

```
def unsafe_increment(n: int) -> None:
    global counter
    for _ in range(n):
        current = counter                # read
        time.sleep(0)                    # yield: lets another thread run (widens the window)
        counter = current + 1            # write back a stale value

lock = threading.Lock()

def safe_increment(n: int) -> None:
    global counter
    for _ in range(n):
        with lock:                       # only one thread inside at a time
            current = counter
            time.sleep(0)
            counter = current + 1
```

Output (4 threads × 1,000 increments each):

```
unsafe counter (expected 4000): 1000
safe counter   (expected 4000): 4000
```

The unsafe number changes from run to run (other runs printed 2000 and 1031). The `time.sleep(0)` just makes the bug show up every time; without it the same bug still exists, it only appears rarely — which is worse, because it passes your tests and fails in production.

How it works — the synchronization toolbox

Primitive	What it does	Typical use
`Lock`	One holder at a time. Not re-entrant: the same thread acquiring twice blocks itself.	Protect shared state
`RLock`	Re-entrant lock: the owning thread may acquire it again (must release as many times).	Recursive code that re-enters a locked method
`Semaphore(n)`	Up to *n* holders at once.	Limit concurrent DB connections to 10
`Event`	A flag threads can `wait()` on until someone calls `set()`.	"Start now" / "shut down" signal
`Condition`	Wait until a condition becomes true, woken by `notify()`.	Building your own queues
`Barrier(n)`	Blocks until *n* threads arrive, then releases all.	Phased simulations
`queue.Queue`	Thread-safe FIFO with blocking `get`/`put`.	Producer–consumer (next section)

Always use locks through `with lock:` — it releases the lock even if an exception is raised. Manual `acquire()`/`release()` is useful only with a timeout: `if lock.acquire(timeout=1): ...`.

Common mistakes

- Believing "the GIL makes my code thread-safe". The GIL protects the *interpreter's* internals (so a `list` is never corrupted), not *your* multi-step logic. `if key not in d: d[key] = compute()` can still run twice.
- Protecting writes but not reads of a multi-field invariant (reading `balance` while another thread is half-way through a transfer).

- Holding a lock while doing slow I/O — other threads queue behind it.

In the interview

> **30-second answer:** "A race condition is when the outcome depends on thread timing, usually from an unprotected read-modify-write. Even `x += 1` is several bytecodes, so it's not atomic. I protect shared state with a `threading.Lock` used as a context manager, keep critical sections small, or avoid sharing entirely by passing messages through `queue.Queue`."

- *Is `list.append` thread-safe?* — A single `append` on a built-in list is safe in CPython (with the GIL, and the free-threaded build uses per-object locks for built-ins), but a sequence of operations is not. Don't rely on it for logic.
- *Lock vs RLock?* — RLock lets the same thread re-acquire; Lock would deadlock on itself.

Self-check

1. Why is `counter += 1` unsafe across threads? → It is load, add, store; another thread can run between them.
2. What does `with lock:` guarantee on an exception? → The lock is released.
3. Which primitive limits concurrency to N? → `Semaphore(N)`.

TL;DR

Shared mutable state + threads = races. Protect with `Lock` (via `with`), keep critical sections tiny, or share nothing and communicate through queues.

16.5 Producer–consumer with `queue.Queue`

What it is

Producer–consumer splits work into threads that *make* items and threads that *process* them, connected by a queue. `queue.Queue` is a thread-safe FIFO (first in, first out) with all locking done for you.

Why it matters

It removes most manual locking: threads never touch each other's data, they only `put` and `get`. A **bounded** queue (`maxsize=N`) also gives **backpressure** — a fast producer blocks on `put` instead of filling memory.

How it works

1. `q = queue.Queue(maxsize=5)`.
2. Producers call `q.put(item)` (blocks while full).
3. Consumers call `q.get()` (blocks while empty), process, then `q.task_done()`.
4. To stop, each producer puts a **sentinel** (a special "no more work" value, here `None`); the consumer stops after seeing one sentinel per producer. (Python 3.13 also added `Queue.shutdown()` for the same purpose.)

```
SENTINEL = None

def producer(q, items):
    for item in items:
        q.put(item)                         # blocks if the queue is full (backpressure)
    q.put(SENTINEL)

def consumer(q, out, n_producers):
    finished = 0
    while finished < n_producers:
        item = q.get()                      # blocks until an item is available
        if item is SENTINEL:
            finished += 1
        else:
            out.append(item * item)
        q.task_done()
```

Output:

```
consumer squared 10 items, sorted: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
```

`q.join()` (not used here) blocks until every item put has been matched by a `task_done()` — handy when the main thread wants to wait for "all work processed" without stopping the workers.

Queue class	Order
`queue.Queue`	FIFO
`queue.LifoQueue`	Stack (last in, first out)
`queue.PriorityQueue`	Smallest item first (put `(priority, item)` tuples)
`queue.SimpleQueue`	Unbounded FIFO, no `task_done` / `join`
`multiprocessing.Queue`	Between processes (items are pickled)
`asyncio.Queue`	Between asyncio tasks (not thread-safe)

Real-world examples

Web crawlers (fetcher threads → parser threads), logging (`logging.handlers.QueueHandler` + `QueueListener` move slow log I/O off request threads), and PyTorch's `DataLoader`, which uses worker *processes* feeding batches through queues.

TL;DR

Share a queue, not variables. Bound it for backpressure. Stop with sentinels or `shutdown()`.

16.6 The GIL (Global Interpreter Lock)

What it is

In the standard CPython build, the **GIL** is a single lock that a thread must hold to execute Python bytecode. So in one process, **only one thread runs Python code at any instant**, even on a 64-core machine.

Why it exists

CPython manages memory with **reference counts** (every object counts how many references point to it; Ch 17). Updating those counts from several threads at once would need a lock on every object or atomic operations everywhere, which was slower and much harder to get right. One big lock made the interpreter simple, single-threaded code fast, and C extensions easy to write — which is a big reason numpy, and the rest of the scientific stack, exist.

How it works

1. A thread holds the GIL while it runs bytecode.
2. It **releases** the GIL when it does blocking I/O (`socket.recv`, file reads, `time.sleep`) and when a C extension explicitly releases it (numpy on large arrays, `hashlib` on big buffers, `zlib`, many image and DB libraries).
3. A thread that wants the GIL waits; after the **switch interval** (`sys.getswitchinterval()`, default 0.005 s = 5 ms) the holder is asked to drop it so others get a turn.

Consequences (the benchmark in 16.2):

Workload	Threads with the GIL
I/O-bound	Scale well — the GIL is released while waiting
CPU-bound pure Python	No speed-up; can even be slightly slower (switching overhead)
CPU-bound in numpy / C that releases the GIL	Can scale across cores

Free-threaded CPython (PEP 703) — status

- **Python 3.13 (Oct 2024):** an *optional, experimental* build with the GIL disabled ships (installers call it "free-threaded"; the executable is `python3.13t`). PEP 703 is the design.
- **Python 3.14 (Oct 2025):** per **PEP 779**, the free-threaded build is **officially supported** (no longer experimental) but **still optional — not the default**. `python3.14` still has a GIL; `python3.14t` does not. The docs put the single-threaded slowdown at roughly 5–10 % (depending on platform and compiler), and memory use is higher.
- **Extensions:** a C extension that is not marked free-threading-compatible makes the interpreter **re-enable the GIL** on import (with a warning). You can also force it: `PYTHON_GIL=1` or `python -X gil=1`. Many major packages (numpy among them) now publish free-threaded wheels, but check before you rely on it.
- **Check at runtime:**

```
import sys, sysconfig
sysconfig.get_config_var("Py_GIL_DISABLED")  # 1 on a free-threaded build, else 0/None
sys._is_gil_enabled()                         # False when running without the GIL (3.13+)
```

- **What changes for you:** on a free-threaded build, the CPU-bound thread benchmark above can scale across cores — but races become *more* likely, not less, so all the locking rules of 16.4 still apply. Built-in `list` / `dict` / `set` use internal per-object locks so they are not corrupted, but multi-step logic still needs your locks.

Other ways around the GIL: **processes** (each has its own GIL), **subinterpreters** — since 3.12 each interpreter can have its own GIL (PEP 684) and 3.14 adds the `concurrent.interpreters` module and `InterpreterPoolExecutor` (PEP 734) — and native code (numpy, Cython, Rust via PyO3) that releases the GIL.

Common mistakes

- "Python can't do concurrency." It does I/O concurrency very well; the GIL limits *parallel Python bytecode*.

- "The GIL makes code thread-safe." See 16.4.
- Assuming other implementations behave the same: Jython and IronPython have no GIL; PyPy has one.

In the interview

> **30-second answer:** "The GIL is a mutex in CPython that allows only one thread to execute Python bytecode at a time. It exists because CPython's reference counting isn't thread-safe without it, and it keeps single-threaded code and C extensions simple. It's released during blocking I/O and by many C extensions, so threads are fine for I/O-bound work; for CPU-bound Python I use multiprocessing or push the work into numpy. Since 3.13 there's an optional free-threaded build, officially supported in 3.14 but not the default."

- *Does the GIL make `x += 1` atomic?* — No. The thread can be switched between the load and the store.
- *Why doesn't numpy suffer from the GIL?* — Its heavy loops run in C and release the GIL.
- *Should I switch to the free-threaded build today?* — Only after checking that all your C-extension dependencies support it and benchmarking; single-threaded code gets a bit slower.

Self-check

1. Default switch interval? → 5 ms.
2. Name two ways to get CPU parallelism in CPython with a GIL. → Processes; C extensions that release the GIL (numpy).
3. How do you tell if the GIL is on at runtime? → `sys._is_gil_enabled()`.
4. Is the free-threaded build the default in 3.14? → No — supported but optional (`python3.14t`).

TL;DR

One thread runs Python bytecode at a time in standard CPython. I/O releases it; pure-Python CPU work doesn't. Free-threading: experimental in 3.13, officially supported but optional in 3.14.

16.7 `multiprocessing`

What it is

`multiprocessing` runs functions in separate **processes**. Each process has its own interpreter, memory and GIL, so CPU-bound Python code runs truly in parallel. The API mirrors `threading`.

How it works

1. **Start method** — how the child process is created:

Method	How	Default on
`spawn`	Fresh interpreter; re-imports your main module; arguments are pickled	Windows, macOS
`fork`	Copies the parent process (fast, but unsafe if the parent has threads)	— (was Linux's default up to 3.13)
`forkserver`	Forks from a clean single-threaded server process	Linux and other POSIX except macOS, from **3.14**

Because `spawn`/`forkserver` re-import your script, **always guard the start-up code with `if __name__ == "__main__":`**, or each child would try to start children of its own.

2. **Pickling** — arguments and return values travel between processes by `pickle` (serialization). Lambdas, nested functions and open sockets can't be pickled; use top-level functions.
3. **No shared memory by default** — each process has its own copy of globals.

```python
counter = 0                              # each process gets its OWN copy

def bump_global(n: int, parent_pid: int) -> None:
    global counter
    for _ in range(n):
        counter += 1
    print(f"  child: different pid from parent={os.getpid() != parent_pid}, sees counter={counter}")

def bump_shared(shared, n: int) -> None:
    for _ in range(n):
        with shared.get_lock():          # Value comes with its own lock
            shared.value += 1

if __name__ == "__main__":
    p = mp.Process(target=bump_global, args=(1000, os.getpid()))
    p.start(); p.join()
    shared = mp.Value("i", 0)            # 'i' = C int in shared memory
    # ... start 4 processes running bump_shared(shared, 1000) ...
    with mp.Pool(processes=4) as pool:
        pool.map(square, range(10))
```

Output (`multiprocessing_demo.py`):

```
start method: spawn, CPUs: 14
  child: different pid from parent=True, sees counter=1000
parent counter after child ran: 0  (not shared!)  exitcode=0
shared Value after 4 x 1000 increments: 4000
pool.map squares: [0, 1, 4, 9, 16, 25, 36, 49, 64, 81]
pool.map chunksize=3: [0, 1, 4, 9, 16] ...
lambda in Pool -> PicklingError
```

Shared-state options and caveats

Tool	What it is	Caveat
Return values / `Queue` / `Pipe`	Message passing (pickled)	Preferred; cost grows with data size
`Value`, `Array`	A C number/array in shared memory	Needs `get_lock()` for read-modify-write
`multiprocessing.shared_memory.SharedMemory`	Raw shared bytes (wrap in a numpy array)	You manage layout, lifetime and `unlink()`
`Manager()`	A server process holding a dict/list proxies talk to	Easy but slow (every access is IPC)

`Pool` methods: `map` (ordered, blocks), `imap`/`imap_unordered` (lazy, stream results), `starmap` (multiple arguments), `apply_async` (one call, returns an `AsyncResult`). The `chunksize` argument batches items per message — large chunks cut overhead for many small tasks.

When not to use it

- Tiny tasks: pickling and process start-up (tens to hundreds of ms under `spawn`) outweigh the work.

- Huge inputs that must be copied to every worker — use shared memory, or let each worker load its own slice from disk.
- I/O-bound work — threads or asyncio are lighter.

Common mistakes

- Missing `if __name__ == "__main__":` → `RuntimeError` or runaway process creation under `spawn`.
- Expecting a global modified in a child to change in the parent.
- Passing lambdas/local functions → `PicklingError` / `AttributeError`.
- Mixing `fork` with threads → child can inherit a lock held by a thread that no longer exists, and hang.

Real-world examples

Gunicorn and uWSGI run several **worker processes** of a Django/Flask app to use all cores; Celery's default worker pool is "prefork" (processes); PyTorch `DataLoader(num_workers=4)` loads batches in worker processes.

TL;DR

Processes = real CPU parallelism at the cost of start-up, pickling and no shared memory. Guard `__main__`, use top-level functions, prefer message passing.

16.8 `concurrent.futures` — executors

What it is

A single high-level API for "run these calls in the background and give me the results": **`ThreadPoolExecutor`** (threads) and **`ProcessPoolExecutor`** (processes) share the same interface. You get a **`Future`** — an object representing a result that will exist later.

How it works

1. `with ThreadPoolExecutor(max_workers=8) as ex:` — creates a pool; leaving the `with` waits for all work (`shutdown(wait=True)`).
2. `fut = ex.submit(fn, *args)` → a `Future`. `fut.result(timeout=None)` waits and returns the value **or re-raises the exception** from the worker.
3. `ex.map(fn, iterable)` → results in **input order**, lazily.
4. `as_completed(futures)` → yields futures in **completion order**.
5. `wait(futures, return_when=FIRST_COMPLETED | FIRST_EXCEPTION | ALL_COMPLETED)` → `(done, pending)`.
6. `fut.cancel()` works only if the call hasn't started yet.

Default pool sizes (3.13+): threads `min(32, os.process_cpu_count() + 4)`; processes `os.process_cpu_count()`.

```python
with ThreadPoolExecutor(max_workers=3) as ex:
    futures = {ex.submit(fake_download, u): u for u in urls}
    for fut in as_completed(futures):
        try:
            print(f"done {futures[fut]} -> {fut.result()}")
        except ConnectionError as e:
            print(f"failed {futures[fut]} -> {e!r}")

with ThreadPoolExecutor(max_workers=3) as ex:
    it = ex.map(fake_download, urls)
    try:
        for r in it:
            print(f"map result {r}")
    except ConnectionError as e:
        print(f"map raised {e!r}")
```

Output (`futures_demo.py`; delays a=0.3 s, b=0.1 s, c=0.2 s and c fails):

```
done b.com -> ('b.com', 100)
failed c.com -> ConnectionError('c.com timed out')
done a.com -> ('a.com', 300)
map result ('a.com', 300)
map result ('b.com', 100)
map raised ConnectionError('c.com timed out')
first completed: 1 done, 1 cancelled
primes: [True, True, True, False] in 0.32s with processes
```

What it shows: `as_completed` hands back the fastest first; `map` keeps input order and raises when it *reaches* the failed item; with one worker and three 0.2 s jobs, after the first finishes the second is already running (can't be cancelled) and only the third is cancelled; switching to processes for CPU work is a one-word change.

Trade-offs

	Raw `threading` / `multiprocessing`	`concurrent.futures`
Return values & exceptions	Manual (queues, shared dicts)	Built in (`Future.result()`)
Thread ↔ process switch	Rewrite	Change the class name
Fine control (daemons, custom sync)	Yes	Limited
Best for	Long-lived workers, pipelines	"Run N calls, collect results"

In the interview

30-second answer: "`concurrent.futures` gives one API over thread and process pools. `submit` returns a Future; `result()` returns the value or re-raises the worker's exception. `map` preserves order, `as_completed` yields in finish order. I use `ThreadPoolExecutor` for blocking I/O and `ProcessPoolExecutor` for CPU-bound pure Python."

- *What happens to an exception in a worker?* — Stored in the Future; raised when you call `result()` (or iterate `map`). If you never check, it is silently lost.
- *How do you add a timeout?* — `fut.result(timeout=5)` or `as_completed(fs, timeout=5)` → `TimeoutError`.

TL;DR

Default choice for simple parallel work. Always call `result()` so exceptions surface.

16.9 `asyncio` — async/await

What it is

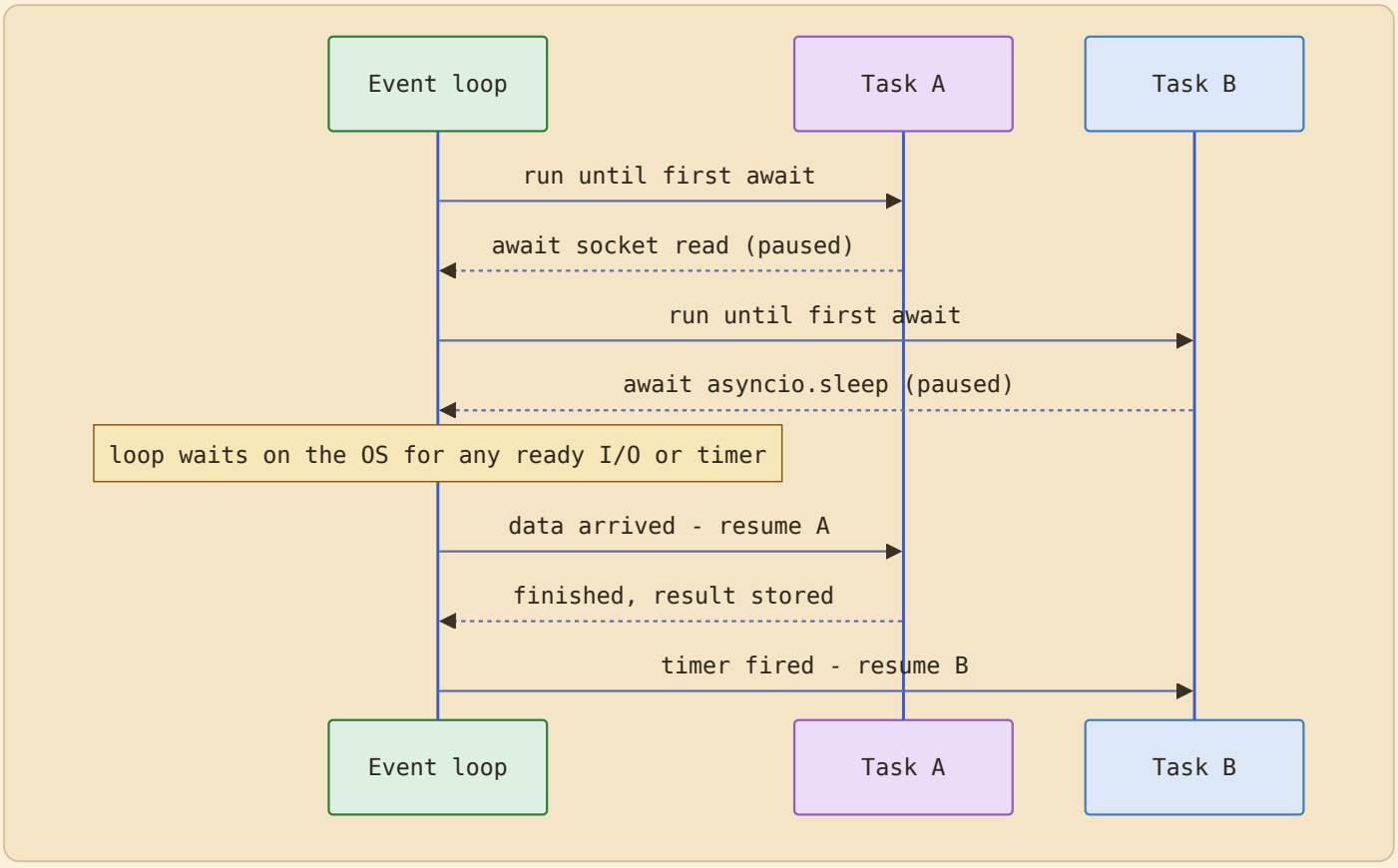
`asyncio` runs many tasks concurrently in **one thread** using an **event loop**. A **coroutine** (`async def`) is a function that can pause at each `await` and let the loop run something else while it waits for I/O. Nothing runs in parallel; the speed comes from never sitting idle.

Analogy: one waiter serving 20 tables. They take an order, hand it to the kitchen (`await`), and serve other tables instead of standing at the kitchen door.

Why it matters

A thread costs memory (its stack) and OS scheduling; 10,000 threads is painful. 10,000 asyncio tasks waiting on sockets is normal. Switching only happens at `await`, so there are far fewer race windows than with threads.

How it works



1. `asyncio.run(main())` creates an event loop, runs `main()` to completion and closes the loop. Call it once, at the top.
2. Calling `fetch()` creates a **coroutine object**; nothing runs until it is awaited or wrapped in a task.
3. `await x` pauses the current coroutine until `x` (a coroutine, Task or Future) is done.
4. `asyncio.create_task(coro)` schedules a coroutine to run **concurrently**, returning a `Task`.
5. The loop uses the OS's readiness APIs (epoll/kqueue) to know which sockets and timers are ready and resumes the corresponding tasks.

Core API with real output (`asyncio_demo.py`)

```python
async def fetch(name: str, delay: float) -> str:
    await asyncio.sleep(delay)           # non-blocking wait
    return f"{name}:{delay}"

await fetch("a", 0.2); await fetch("b", 0.2)                        # sequential: 0.4 s
results = await asyncio.gather(fetch("a", 0.2), fetch("b", 0.2), fetch("c", 0.1))  # 0.2 s

task = asyncio.create_task(fetch("bg", 0.2))    # starts running now
await asyncio.sleep(0.1)
print(task.done())                              # False
value = await task
```

```
type before await: coroutine
awaited: x:0
[0.40s] two sequential awaits
[0.20s] gather -> ['a:0.2', 'b:0.2', 'c:0.1']  (order = argument order)
[0.10s] task done yet? False
[0.20s] task result bg:0.2
```

Key lesson: `await a(); await b()` is **sequential**. Concurrency only happens with `gather`, `create_task` or a `TaskGroup`.

`gather` vs `TaskGroup` (3.11+)

```python
res = await asyncio.gather(fetch("ok", 0.1), failing(0.05), return_exceptions=True)

try:
    async with asyncio.TaskGroup() as tg:
        slow = tg.create_task(fetch("slow", 1.0))
        tg.create_task(failing(0.1))
except* ValueError as eg:                       # except* handles an ExceptionGroup
    print(eg.exceptions, slow.cancelled())
```

```
gather with return_exceptions -> ['ok:0.1', ValueError('boom')]
[0.10s] TaskGroup raised (ValueError('boom'),); slow cancelled=True
```

	`asyncio.gather`	`asyncio.TaskGroup`
On first failure	Raises it, but the **other tasks keep running** unattended	**Cancels** the siblings, waits for them, raises an `ExceptionGroup`
Collect errors as values	`return_exceptions=True`	Catch with `except*`
Results	List in argument order	`task.result()` on each task
Recommended for new code	Simple fan-out	Yes — "structured concurrency": no task outlives its block

Timeouts, limits and async iteration

```python
async with asyncio.timeout(0.1):              # 3.11+, can wrap several awaits
    await fetch("too-slow", 1.0)
await asyncio.wait_for(fetch("too-slow", 1.0), timeout=0.1)   # one awaitable

sem = asyncio.Semaphore(2)                    # at most 2 at a time
async def limited(i):
    async with sem:
        await asyncio.sleep(0.1)
        return i

async def ticker(n):                          # async generator
    for i in range(n):
        await asyncio.sleep(0.05)
        yield i

async with AsyncResource():                   # __aenter__ / __aexit__
    squares = [i * i async for i in ticker(4)] # async comprehension
```

```
asyncio.timeout -> TimeoutError
wait_for -> TimeoutError
[0.31s] 6 jobs, 2 at a time -> [0, 1, 2, 3, 4, 5]
[0.00s] open resource
[0.20s] close resource
async comprehension -> [0, 1, 4, 9]
```

On timeout the inner task is **cancelled** (a `CancelledError` is raised inside it) and the caller sees the built-in `TimeoutError` (since 3.11, `asyncio.TimeoutError` is the same class). Six jobs with a limit of two took 3 rounds × 0.1 s.

Async protocol summary: `async for` uses `__aiter__`/`__anext__`; `async with` uses `__aenter__`/`__aexit__`. Other useful pieces: `asyncio.Queue` (producer–consumer between tasks), `asyncio.Event`, `asyncio.Lock`, `asyncio.as_completed`.

Mixing sync and async

```python
def blocking_io() -> str:
    time.sleep(0.3)                  # a sync library call (e.g. requests.get)
    return "sync result"

await asyncio.gather(asyncio.to_thread(blocking_io), asyncio.to_thread(blocking_io))
async def bad():
    return blocking_io()             # time.sleep inside a coroutine blocks EVERYTHING
await asyncio.gather(bad(), bad())
```

```
[0.31s] two blocking calls via to_thread ran together
[0.60s] two blocking calls inside coroutines ran one after another
```

One blocking call inside a coroutine freezes the **entire** loop — every other task, every connection. Use async libraries (`httpx`/`aiohttp` instead of `requests`, `asyncpg` instead of `psycopg2` sync, `asyncio.sleep` instead of `time.sleep`), or push sync calls to a thread with `asyncio.to_thread` (3.9+) / `loop.run_in_executor`. CPU-heavy work goes to a `ProcessPoolExecutor` via `run_in_executor`.

When to use asyncio

- **Yes:** many simultaneous network connections — web servers/APIs (FastAPI, Starlette, aiohttp), websockets, chat, crawlers, calling many microservices or LLM APIs concurrently.
- **No:** CPU-bound work (single thread!), a script that makes three HTTP calls (threads are simpler), or a codebase whose key libraries are all synchronous.

Common mistakes

- Forgetting `await`: `fetch()` without `await` just creates a coroutine; Python warns *"coroutine ... was never awaited"*.
- `create_task` and dropping the reference: the loop keeps only a weak reference, so an unreferenced task can be garbage-collected mid-run. Keep tasks in a set or use a `TaskGroup`.
- Calling `asyncio.run()` inside an already-running loop (e.g. Jupyter) → `RuntimeError`; in notebooks just `await main()`.
- Catching `CancelledError` and not re-raising it — breaks timeouts and TaskGroup cancellation.
- Sharing an `asyncio.Queue`/`Lock` with other threads — asyncio objects are not thread-safe; use `loop.call_soon_threadsafe` or `asyncio.run_coroutine_threadsafe`.

Real-world examples

FastAPI/Starlette serve async endpoints on the uvicorn server (uvloop, a faster event loop written on libuv, is optional); Discord bots (`discord.py`), Home Assistant, and the official Anthropic and OpenAI Python SDKs ship `Async...` clients for concurrent requests.

In the interview

> **30-second answer:** "asyncio is single-threaded cooperative concurrency. Coroutines pause at `await`, and the event loop resumes whichever task's I/O is ready. It scales to thousands of connections cheaply, but anything blocking stalls everything, so I use async libraries or `asyncio.to_thread`. For fan-out I prefer `TaskGroup` over `gather` because a failure cancels the siblings."

- *Is asyncio parallel?* — No, one thread; concurrency only.
- *`gather` vs `TaskGroup`?* — `gather` leaves siblings running on error unless you handle it; `TaskGroup` cancels them and raises an `ExceptionGroup`.
- *How do you call sync code from async?* — `await asyncio.to_thread(fn, *args)`.

Self-check

1. What runs when you call an `async def` function? → Nothing; you get a coroutine object.
2. Two `await`s in a row — concurrent? → No, sequential.
3. How do you cap concurrency at 10? → `asyncio.Semaphore(10)`.
4. What exception does `asyncio.timeout` raise to the caller? → `TimeoutError`.
5. Why is `time.sleep` in a coroutine bad? → It blocks the whole event loop.

TL;DR

One thread, one loop, many tasks; switch only at `await`. `asyncio.run` once, `TaskGroup`/`gather` for concurrency, `timeout` for deadlines, `Semaphore` for limits, `to_thread` for sync calls. Never block the loop.

16.10 Choosing the right tool

Situation	Use	Why
A few to hundreds of blocking I/O calls, sync libraries	`ThreadPoolExecutor`	Simple, GIL released during I/O
Thousands of concurrent connections, async libraries	`asyncio`	Tasks are cheap, no thread per connection
CPU-bound pure Python	`ProcessPoolExecutor` / `multiprocessing.Pool`	Each process has its own GIL
CPU-bound numeric work	numpy/pandas (vectorize, Ch 17–18)	C loops, often multi-threaded internally
CPU-bound, want shared memory, deps support it	Free-threaded build (3.14t) + threads	Real parallel threads
Long-running pipeline stages	Threads/processes + `Queue`	Backpressure, clean shutdown
Async app needs a sync library	`asyncio.to_thread`	Keeps the loop responsive
Async app needs heavy CPU	`loop.run_in_executor(ProcessPoolExecutor(), ...)`	Moves CPU work off the loop
Scale beyond one machine	Celery, RQ, Ray, Dask, Spark	Distributed task queues/clusters

	Threads	Processes	asyncio
Parallel CPU (GIL build)	No	Yes	No
Memory per unit	~MB stack (virtual)	Whole interpreter (tens of MB)	~KB per task
Shared memory	Yes (needs locks)	No (pickle/IPC)	Yes (single thread)
Switching	Pre-emptive (any time)	OS	Cooperative (at `await`)
Race risk	High	Low	Low
Works with sync libs	Yes	Yes	Needs `to_thread`

16.11 Common pitfalls

Blocking calls inside async code

Shown in 16.9: `time.sleep`, `requests.get`, a sync DB driver, big file reads, or a long CPU loop in a coroutine halts every other task. Diagnose with **debug mode** (`asyncio.run(main(), debug=True)` or `PYTHONASYNCIODEBUG=1`), which logs callbacks slower than 100 ms. Python 3.14 also adds `python -m asyncio ps PID` / `pstree PID` to inspect the tasks of a running process.

Deadlocks

A **deadlock** happens when thread 1 holds lock A and waits for B, while thread 2 holds B and waits for A. Both wait forever.

```
def transfer(first, second, name, log):
    with first:
        time.sleep(0.05)                              # make the bad interleaving likely
        if second.acquire(timeout=0.5):               # a real deadlock would wait forever
            second.release()
            log.append(f"{name} ok")
        else:
            log.append(f"{name} gave up: DEADLOCK")

run((lock_a, lock_b), (lock_b, lock_a))   # opposite order
run((lock_a, lock_b), (lock_a, lock_b))   # same order
```

Output (`pitfalls_demo.py`):

```
opposite order : ['T1 gave up: DEADLOCK', 'T2 ok']
same order     : ['T1 ok', 'T2 ok']
RLock re-entered by the same thread: fine
plain Lock re-acquire (timeout 0.1): False
```

The timeout lets the demo finish; without it both threads would hang. Fixes:

1. **Global lock ordering** — always acquire locks in the same order (e.g. by account id).
2. **Fewer locks** — one coarse lock, or no shared state (queues, immutable data).
3. **Timeouts** on `acquire` to detect and back off.
4. `RLock` when the same thread legitimately re-enters (a plain `Lock` deadlocks on itself, as the last line shows).
5. In asyncio: a task awaiting something that only it could produce, or a TaskGroup child waiting on its parent — same idea, same cure (clear ownership).

Other pitfalls

Pitfall	Symptom	Fix
Unprotected shared state	Wrong totals, rare	Lock, or message passing
Unhandled exception in a `Thread`	Printed to stderr, thread dies silently	Use executors and call `result()`; or set `threading.excepthook`
Too many threads	Memory, context-switch overhead	Bounded pool
Missing `__main__` guard	Recursive spawning / `RuntimeError`	Guard the entry point
Huge arguments to processes	Slow pickling	Shared memory, or load inside the worker
Forgotten `await` / lost task reference	Work silently never runs / vanishes	Linters (ruff), `TaskGroup`
`fork` after threads started	Random hangs in child	`spawn` / `forkserver` (defaults on macOS/Windows, and Linux from 3.14)

In the interview

- *Four conditions for deadlock?* — Mutual exclusion, hold-and-wait, no pre-emption, circular wait. Break any one (lock ordering breaks circular wait). Details: LLD Ch 12.
- *How would you find a hung thread?* — `faulthandler.dump_traceback_later()` or `py-spy dump --pid` show every thread's stack.

16.12 Chapter self-check

1. Concurrency vs parallelism in one line each? → Overlapping waits vs simultaneous execution on several cores.
2. Why do threads speed up downloads but not a pure-Python prime search? → I/O releases the GIL; bytecode execution doesn't.
3. What does `q.task_done()` pair with? → `q.join()`, which waits until every item is marked done.
4. Why must `multiprocessing` code have `if __name__ == "__main__":`? → `spawn`/`forkserver` re-import the main module in each child.
5. Which default start method does Linux use from 3.14? → `forkserver`.
6. `ex.map` vs `as_completed` ordering? → Input order vs completion order.
7. What happens to siblings when one TaskGroup task fails? → They are cancelled; an `ExceptionGroup` is raised.
8. How do you run `requests.get` from async code without freezing the loop? → `await asyncio.to_thread(requests.get, url)`.
9. What's the status of free-threaded CPython in 3.14? → Officially supported (PEP 779), optional, not the default.
10. The simplest way to prevent lock-ordering deadlocks? → Always acquire locks in one global order.

Chapter TL;DR

- **First classify the work**: waiting (I/O-bound) or computing (CPU-bound). Measure.
- **Threads**: cheap concurrency for I/O; share memory → protect with `Lock`, or pass messages through `queue.Queue` (bounded, with sentinels).
- **GIL**: one thread runs Python bytecode at a time in standard CPython; released on I/O and by C extensions. Free-threaded build: experimental in 3.13, supported but optional in 3.14 (`python3.14t`, ~5–10 % single-thread cost).
- **Processes**: real parallelism; no shared globals; arguments are pickled; guard `__main__`; `spawn` on macOS/Windows, `forkserver` on Linux from 3.14.
- **Executors**: `ThreadPoolExecutor`/`ProcessPoolExecutor`, `submit` → `Future.result()`, `map` (ordered), `as_completed` (finish order). The default choice for "run N things".
- **asyncio**: one thread, event loop, coroutines pause at `await`. `gather`/`TaskGroup` for concurrency, `asyncio.timeout`, `Semaphore`, async generators/context managers, `to_thread` for sync code. Never block the loop.
- **Pitfalls**: races, deadlocks (fix with lock ordering), blocking calls in async, lost exceptions, missing `__main__` guard.